

**Name:** K RAJ BHARATH K RAJ BHARATH | **Reg. No:** RA2511003011548

## Unit – 3 : Session – 3 : SLO – 2 | Synchronization and Inter-Thread Communication

### ■ Worksheet (Answers)

- Fill: The synchronized keyword is used to prevent RACE conditions between threads.
- T/F: A synchronized block can only lock on this object. — FALSE (it can lock on any object reference)
- MCQ: Method used for thread communication → A) notify()
- Fill: The wait() method should be called inside a SYNCHRONIZED block or method.
- Find the output: synchronized increment() called 2000 times total across two threads with join → A) 2000
- T/F: notifyAll() wakes up all threads waiting on the object's monitor. — TRUE
- MCQ: wait() throws an exception when → D) Both A and C (called outside synchronized context → IllegalMonitorStateException; thread interrupted → InterruptedException)
- Fill: The notify() method wakes up one thread waiting on the OBJECT'S MONITOR.
- Find the error: lock.wait() outside synchronized → C) IllegalMonitorStateException
- Fill: The synchronized keyword can be applied to methods and BLOCKS.